

BAAZ & UDAY JKBOSE

2026 CLASS 11TH

OSCILLATIONS

LECTURE-03 **PHYSICS**

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Topics To Be Covered

- 1 Revision dec-1
dec-2
- 2 Energy in SHM
- 3 K.E
- 4 P.E
- 5 Total Energy

lecture no. 3

2026/2025/2023/ (3m & 5m)

Energy in SHM

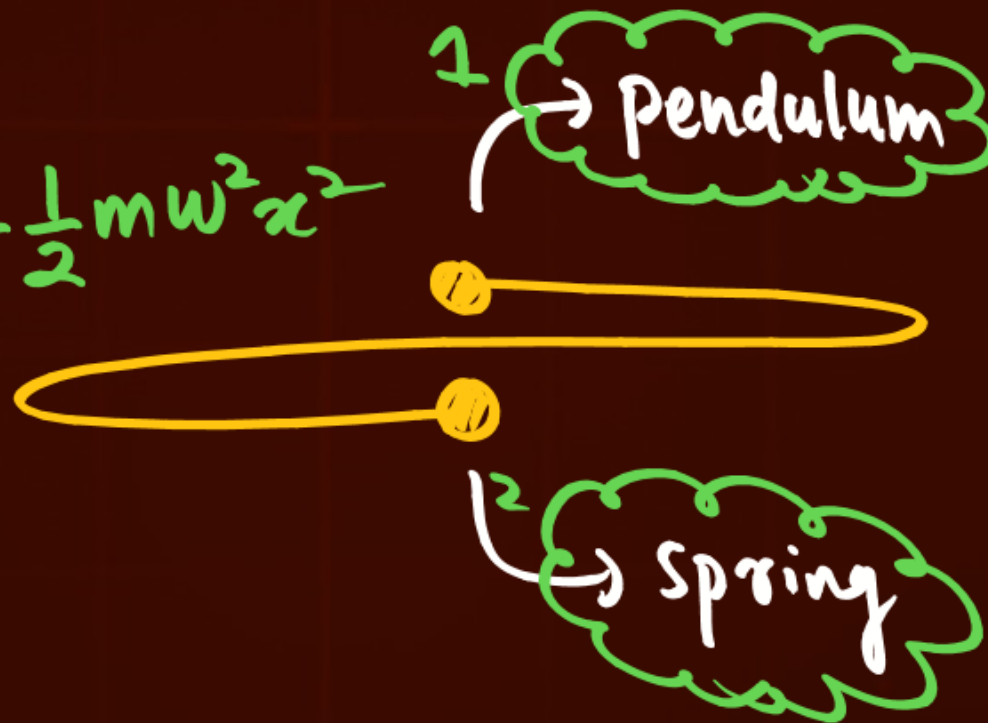
①

Energy is the capacity to do work

$$T.E = K.E + P.E$$

$$\checkmark \underline{K.E} = \frac{1}{2}mv^2 = \frac{1}{2}m\omega^2 A^2 - \frac{1}{2}m\omega^2 x^2$$

$$\checkmark P.E = ?$$



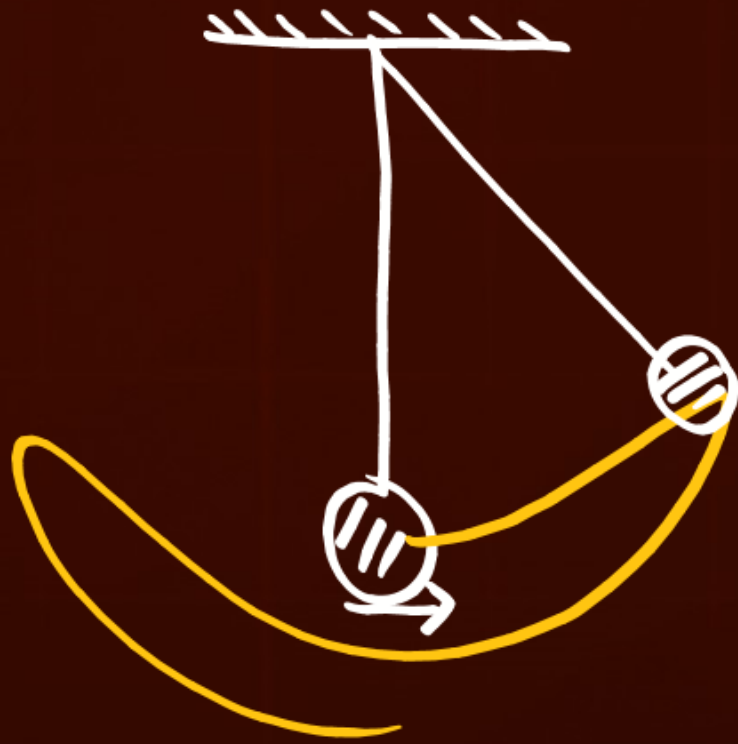
# K.E	# T.E
# P.E	

(1) Kinetic Energy in SHM

the energy by which Body moves

$$K.E = \frac{1}{2} m v^2$$

$$v = A\omega \cos(\omega t)$$



$$2 \times 4 = 8$$
$$4 \times 2 = 8$$

$$\# K.E = \frac{1}{2} m v^2$$

$$v = A\omega \cos(\omega t)$$

$$K.E = \frac{1}{2} m (A\omega \cos(\omega t))^2$$

$$K.E = \frac{1}{2} m (A^2 \omega^2 \cos^2 \omega t)$$

$$K.E = \frac{1}{2} m A^2 \omega^2 \cos^2 \omega t$$

$$\cos^2 \omega t = 1 - \sin^2 \omega t$$

$$K.E = \frac{1}{2} m A^2 \omega^2 (1 - \sin^2 \omega t)$$

$$K.E = \frac{1}{2} m \underbrace{A^2 \omega^2}_{\chi^2} - \frac{1}{2} m \underbrace{A^2 \omega^2}_{\chi^2} \sin^2 \omega t$$

$$K.E = \frac{1}{2} m \omega^2 A^2 - \frac{1}{2} m \omega^2 A^2 \sin^2 \omega t$$

$$K.E = \frac{1}{2} m \omega^2 A^2 - \frac{1}{2} m \omega^2 \chi^2$$

$$\# \quad \omega = \frac{\theta}{t}$$

$$\underline{\omega t = \theta}$$

$$\# \quad (\cos \theta)^2 = \cos^2 \theta$$

$$\# \quad (\cos \omega t)^2 = \cos^2 \omega t$$

$$\# \quad \sin^2 \theta + \cos^2 \theta = 1$$

$$\sin^2 \omega t + \cos^2 \omega t = 1$$

$$\cos^2 \omega t = 1 - \sin^2 \omega t$$

$$\# \quad x = A \sin \omega t$$

$$x^2 = (A \sin \omega t)^2$$

$$\underline{x^2 = A^2 \sin^2 \omega t}$$

Feel $k \cdot x$

2024 (MCQ)

① $k \cdot x = \frac{1}{2} m \omega^2 A^2 - \frac{1}{2} m \omega^2 x^2$ ↑

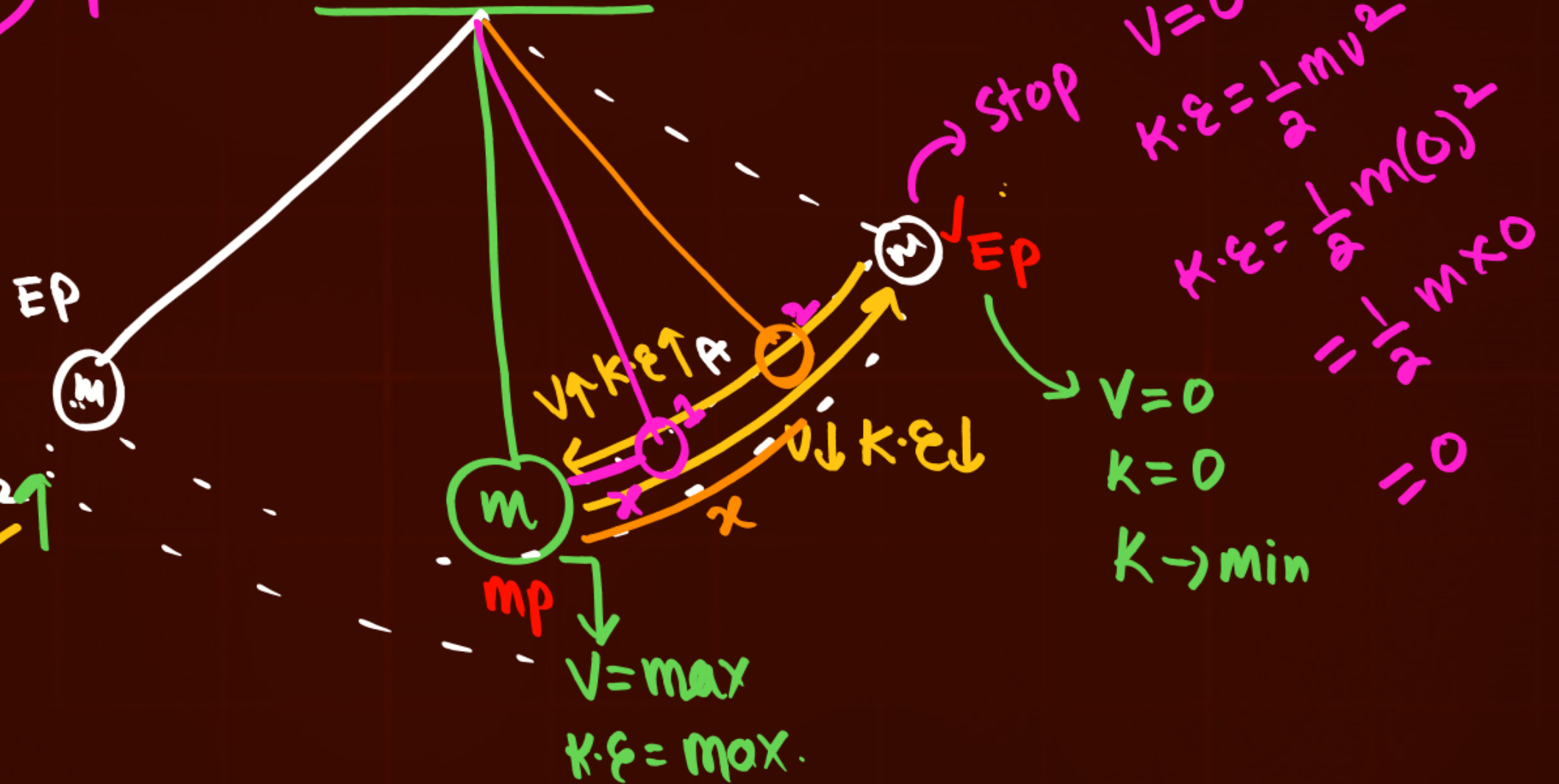
$k \cdot x = 10 - 4$

$k \cdot x = 6 \text{ J}$

② $k \cdot x = \frac{1}{2} m \omega^2 A^2 - \frac{1}{2} m \omega^2 x^2$ ↑

$k \cdot x = 10 - 6$

$k \cdot x = 4$



Graph of K.E with x



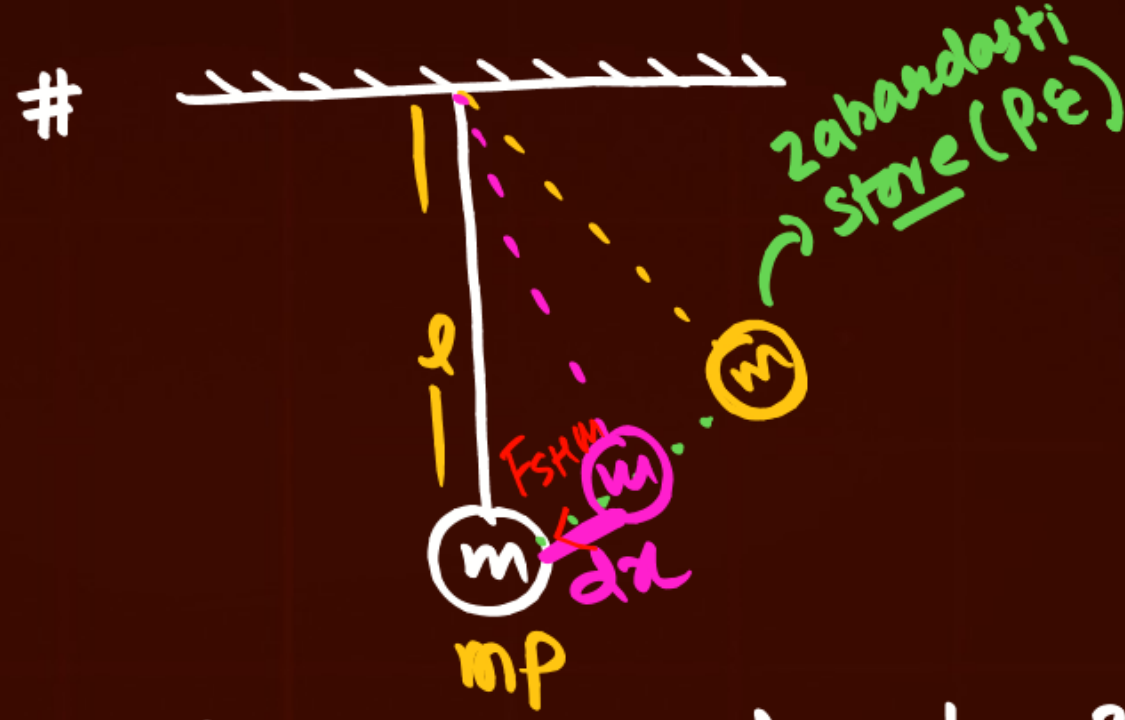
② P.E in SHM

The Energy Store in a Body due to Relative position of particle

Zabardasti ✓

P.E = Work done

↓
F_{SHM}



P.E = work done by SHM Force

$w = Fx$

$dw = F dx$

$F_{SHM} = m\omega^2 x$
↑
Variable

$dw = m\omega^2 x dx$

on integration b/w

$\int dw = \int m\omega^2 x dx$

$$\int \underline{dw} = \int \underline{m\omega^2 x} dx$$

$$\underline{\text{Work}} = m\omega^2 \int x dx$$

$$W = m\omega^2 \frac{x^2}{2}$$

$$W = \frac{m\omega^2 x^2}{2}$$

$$W = \frac{1}{2} m\omega^2 x^2$$

$$\int x' dx = \frac{x^{1+1}}{1+1} = \frac{x^2}{2}$$

$$W = \frac{1}{2} m\omega^2 x^2$$

$$P.E = \frac{1}{2} m\omega^2 x^2$$

$$U = \frac{1}{2} m\omega^2 x^2$$

Feel - PE



$$\uparrow \underline{P.E} = \frac{1}{2} m \omega^2 \underline{x^2} \uparrow$$

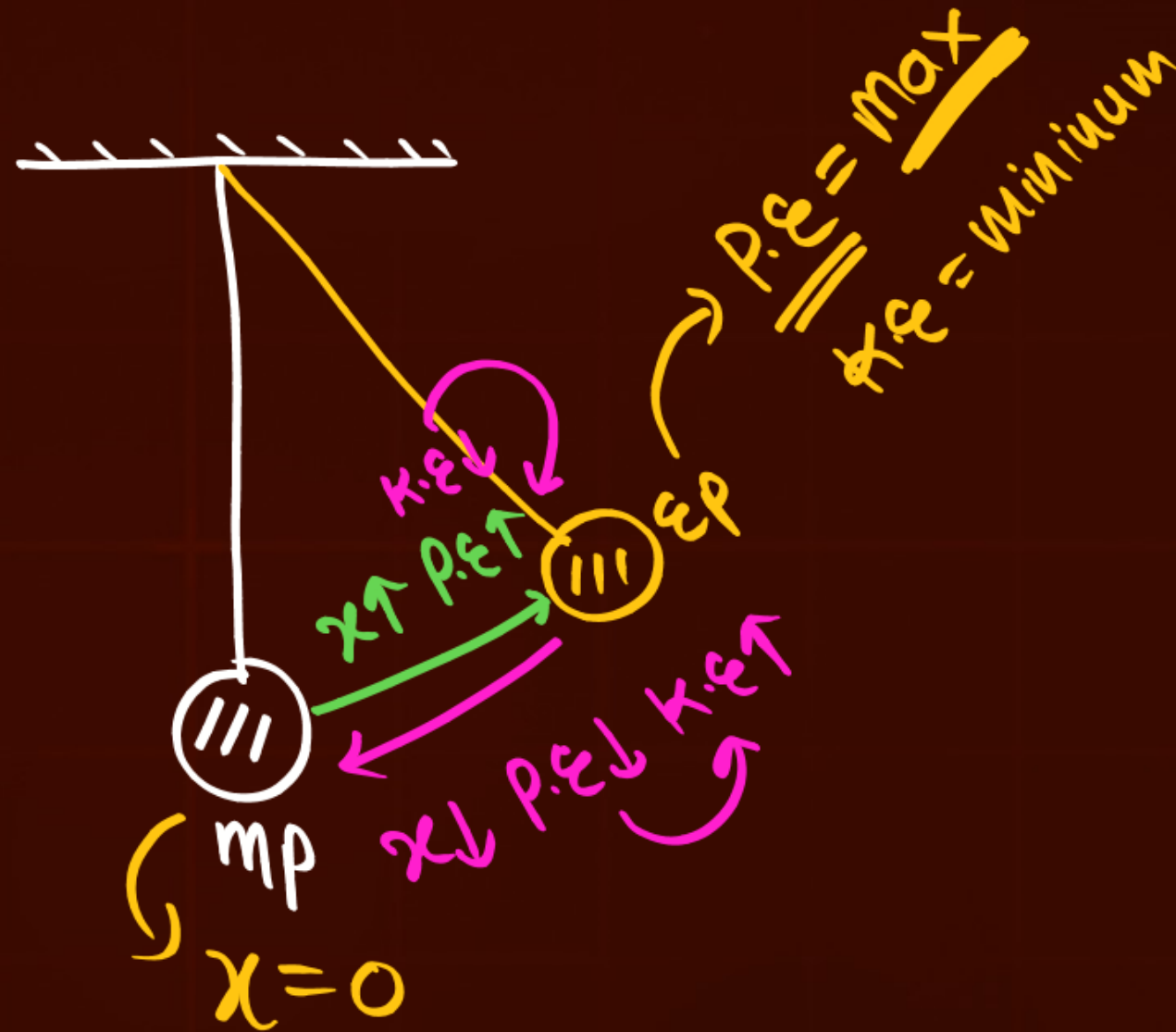
At EP $(x=A)$ (max)

$$P.E = \frac{1}{2} m \omega^2 A^2 \text{ (Max)}$$

At MP $(x=0)$ (Minimum)

$$P.E = \frac{1}{2} m \omega^2 0^2$$

$$P.E = \frac{1}{2} m \omega^2 x_0 = 0$$

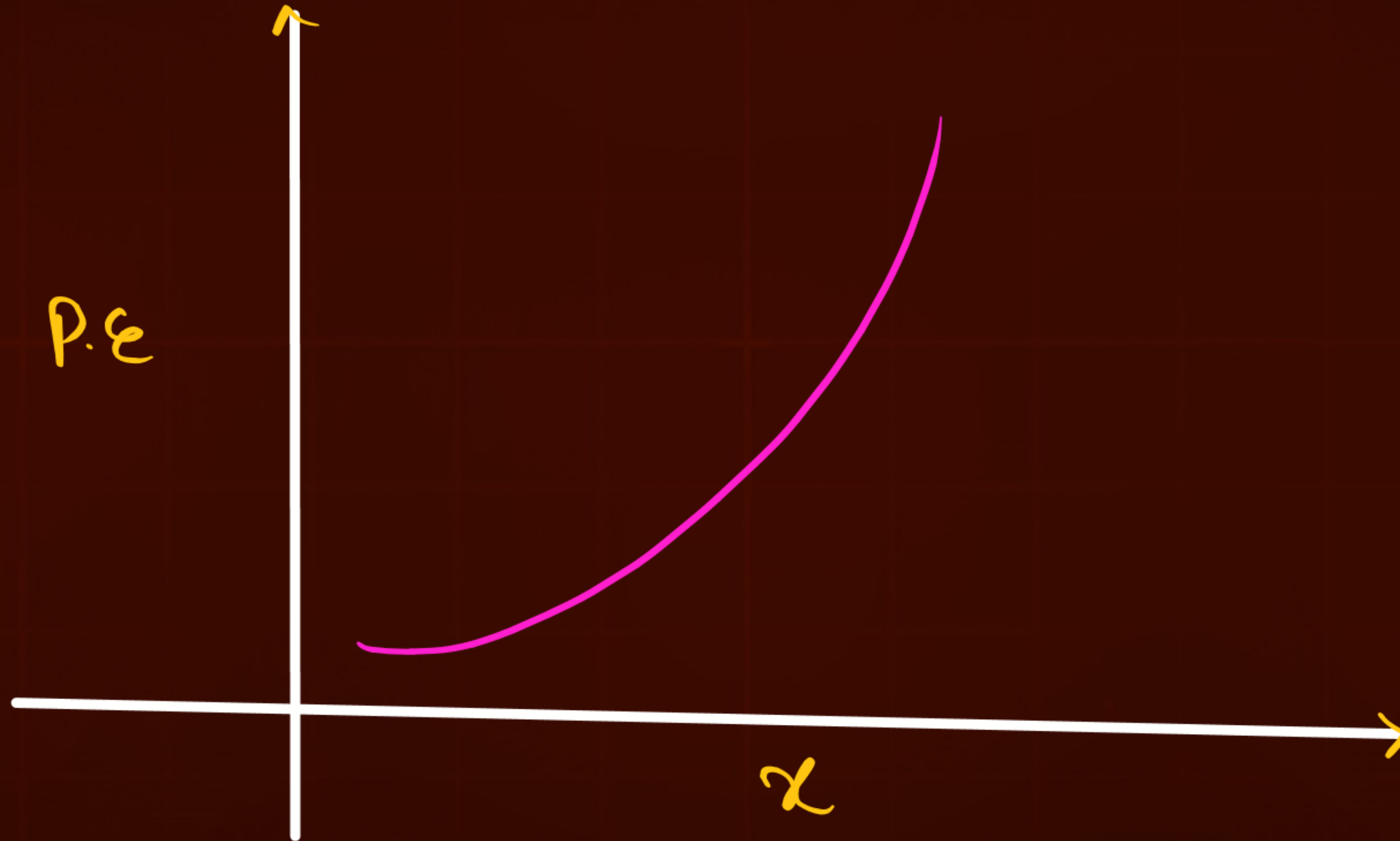


P.E (Mini)

$$P.E = \frac{1}{2} m \omega^2 x^2$$

$$P.E = \frac{1}{2} m \omega^2 0^2 = 0$$

Graph of P.E with (x)



(13) Total Energy (T.E) / Total Mechanical Energy

It is the Sum of K.E + P.E

$T.E = \underline{K.E} + P.E$ ✓

$$T.E = \left(\frac{1}{2} m \omega^2 A^2 - \frac{1}{2} m \omega^2 x^2 \right) + \frac{1}{2} m \omega^2 x^2$$

$$\boxed{T.E = \frac{1}{2} m \omega^2 A^2}$$

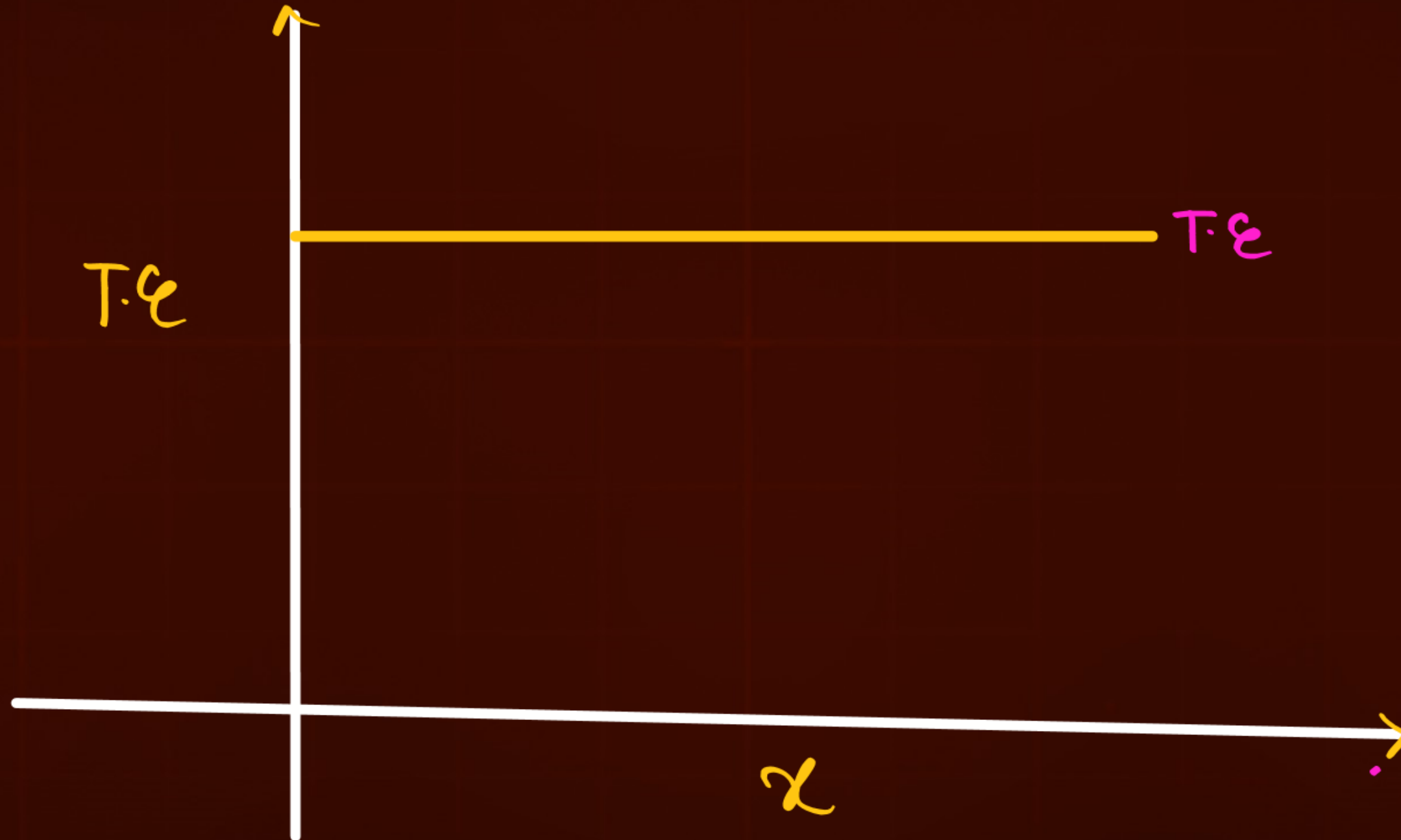
↓ Constant

$$x - x = 0$$

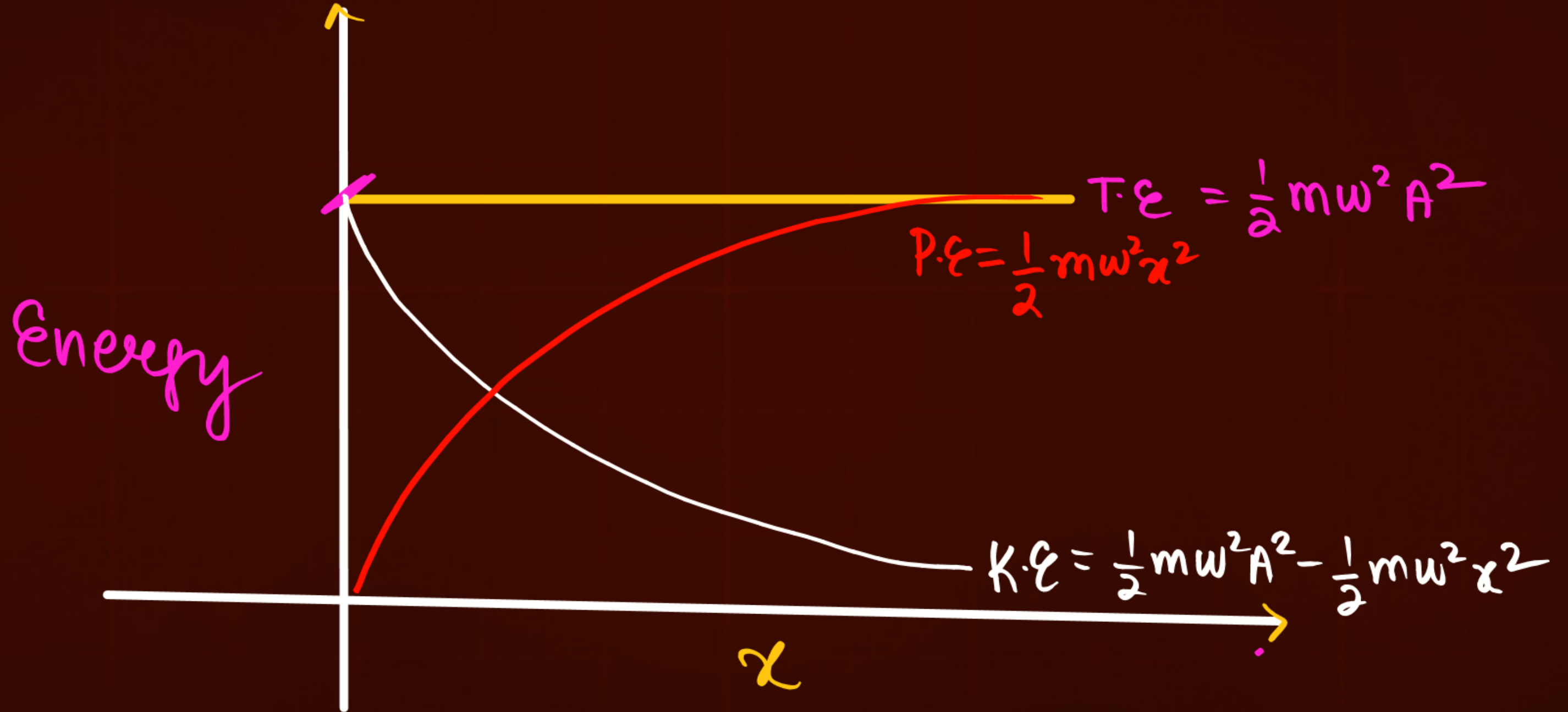
$$x^2 - x^2 = 0$$

$$a^4 - a^4 = 0$$

Graph of T.E with (x)



General Graph of Energy in SHM



SUMMARY

✓E

✓K.E

✓P.E

✓T.E

HOMework



Derivation

Mehnat Karo Mehnat karo
Mehnat karne se ho jata hai
Insallah apke 498 No. Aayege
Bas Yhi motivation apne dil me
rkho bs



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